

Master **systemctl** The Ultimate Linux Service Manager

Struggling with Linux services? systemctl is your Swiss
Army knife for controlling systemd.

Let's break it down!

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What is systemctl?

systemctl is **the primary command-line tool** for systemd (Linux's modern init system).

Manages not just services, but also:

- System states (targets, like runlevels).
- Timers (cron alternatives).
- Mounts, sockets, devices.

Why it matters?

Replaces legacy scripts (/etc/init.d/) with faster, parallelized service starts.

Fact:

Ubuntu, Fedora, and Debian all use systemd by default.

Core Commands

systemctl start apache2

Starts Apache (but won't auto-start on reboot).

systemctl stop mongod

Stops MongoDB (useful for maintenance).

systemctl restart nginx

Restarts NGINX (loads new configs).

systemctl reload sshd

Reloads SSHD (no dropped connections!).

Key Difference:

- **restart** = Hard stop + start.
- **reload** = Graceful config reload (avoids downtime).

Enable/Disable

systemctl enable docker

Starts Docker on every boot.

systemctl disable cups

Stops CUPS (printer service) from auto-starting.

systemctl is-enabled ssh

Checks if SSH starts at boot.

Combine with **--now** to enable AND start immediately:

systemctl enable --now postgresql

Troubleshooting

Check Status:

```
systemctl status mysql
```

Shows live state + recent logs.

View Logs:

```
journalctl -u mysql -n 50
```

Last 50 logs for MySQL.

```
journalctl -xe
```

System-wide errors.

List All Services:

```
systemctl list-units --type=service --all
```

When to use?

- Service fails to start.
- Debugging slow boots.

System Targets & Dependencies

Targets = Modern 'Runlevels'

Graphical Mode: **graphical.target**

CLI Mode: **multi-user.target**

Rescue Mode: **rescue.target**

systemctl isolate multi-user.target

Switch to CLI-only.

systemctl set-default graphical.target

Boot into GUI by default.

systemctl list-dependencies apache2

See what Apache depends on.

systemd Timers (Cron Alternative)

Create a Timer:



```
# /etc/systemd/system/backup.timer  
[Timer]OnCalendar=*-*-* 02:00:00 # Runs daily at 2 AM.  
Unit=backup.service
```

Enable It:

systemctl enable --now backup.timer

Why Timers?

- Logs via journalctl.
- Supports dependencies.